

supplementary File-1

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1 Model strategies, parameter size, performance, and comparability tables

Table 1: Model strategies, parameter size, performance, and comparability level on the Cityscapes dataset

SL	Model name	Characteristics 1	Characteristics 2	Parameters (M)	mIoU	Comp.	Year	Source type	Ref
1	Panoptic-DeepLab	Dual-ASPP and Decoder	Class-Agnostic Object Centre Prediction	43.6	84.2%	H	2020	Published	[1]
2	EfficientPS	New Semantic Head	Panoptic Fusion Module	40.89	84.21%	H	2021	Published	[2]
3	DCNAS	Densely Connected Search Space	Fusion Module for Memory Efficiency	42	84.3%	H	2021	Published	[3]
4	HRNetV2 + OCR	Object-Contextual Representation	Pixel-Object Relation Module	10.5	84.5%	H	2021	Published	[4]
5	Depth Anything	Self-supervised Depth Estimation	Scalable Unsupervised Data Augmentation	335	84.8%	L	2024	Published	[5]
6	SERNet-Former	Efficient ResNet with AbGs and AbMs	Attention-Fusion Networks	42.4	84.82%	H	2024	Published	[6]
7	InverseForm	Inverse Transformation Network for Homography Prediction	Distance-Based Loss Function Incorporating Homography Metrics	~28	85.6%	M	2021	Published	[7]
8	HS3-Fuse	Dynamic Task Complexity Adaptation	Multilabel Hierarchical Supervision with Feature Integration	~	85.8%	M	2021	Preprint	[8]
9	VLTSeg	Adaptive Cross-Domain Contrastive Learning	Vision-Language Guided Segmentation Refinement	304	86.1%	L	2023	Preprint	[9]
10	MetaPrompt-SD	Perceptual Diffusion Model	Text-to-Image Diffusion with Meta-Prompt	912	86.2%	L	2023	Preprint	[10]

Comp. = Comparability level. H (High): evaluated under standard benchmark protocols with no substantial external supervision reported. M (Moderate): includes specialised training, supervision, or evaluation strategies that may affect direct comparison. L (Limited): relies on large-scale pretraining, vision-language supervision, diffusion models, foundation models, or other external data sources, reducing direct comparability with conventional semantic segmentation models. Reported mIoU values are reproduced from the original studies and may not be strictly comparable across different evaluation protocols, training schedules, pretraining datasets, data augmentation strategies, or evaluation settings.

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Table 2: Model strategies, parameter size, performance, and comparability level on the CamVid dataset

SL	Model name	Characteristics 1	Characteristics 2	Parameters (M)	mIoU	Comp.	Year	Source type	Ref
1	EDA-Net	Efficient Dense Module	Asymmetric Convolution	0.68	64.4%	H	2019	Published	[11]
2	FC-DenseNet103	Fully Convolutional DenseNet	Transition Down and Transition Up Layers	9.4	66.9%	H	2017	Published	[12]
3	BiSeNet	Spatial Path	Context Path	49.0	68.7%	M	2018	Published	[13]
4	DenseDecoder	Dense Decoder Shortcut Connections	Single-Pass Segmentation Architecture	~	70.9%	H	2018	Published	[14]
5	VideoGCRF	Deep Spatio-Temporal Random Field	Efficient Inference Mechanisms	~44	75.2%	M	2018	Published	[15]
6	ETC-Mobile	Lightweight DNN for Pre-frame Inference	Spatial Contextual Information Aggregation	3.2	76.3%	M	2020	Published	[16]
7	RTFormer-Base	GPU-Friendly Attention	Cross-Resolution Attention	16.8	82.5%	M	2022	Published	[17]
8	DeepLabV3Plus + SDCNetAug	Optical Flow-Based Video Propagation	Probabilistic Boundary Label Relaxation	~	82.9%	L	2019	Published	[18]
9	SIW	Class Label Embeddings	Data Merging from Comprehensive Training	~	83.7%	L	2024	Published	[19]
10	SERNet-Former	Efficient ResNet with AbGs and AbMs	Attention-Fusion Networks	42.4	84.62%	H	2024	Published	[6]

Comp. = Comparability level. H (High): evaluated under standard CamVid benchmark protocols with no substantial external supervision reported. M (Moderate): includes pretrained backbones, temporal information, specialised supervision, or training strategies that may affect direct comparison. L (Limited): relies on additional training data, video propagation, data merging, large-scale pretraining, or other external resources that reduce direct comparability with conventional semantic segmentation models. Reported mIoU values are reproduced from the original studies and may not be strictly comparable across different evaluation protocols, training schedules, pretraining datasets, image resolutions, data augmentation strategies, or evaluation settings.

Table 3: Model strategies, parameter size, performance, and comparability level on the PASCAL VOC 2012 dataset

SL	Model name	Characteristics 1	Characteristics 2	Parameters (M)	mIoU	Comp.	Year	Source type	Ref
1	HyperSeg-L	Spatially Varying Weights in the Decoder	Efficient Hypernetwork Design	39.6	80.6%	H	2021	Published	[20]
2	Auto-DeepLab-L	Hierarchical Neural Architecture Search	Cell Level Search, Network Level Search	44.4	82.04%	H	2019	Published	[21]
3	ResNet-GCN	Global Convolutional Network	Residual-based Boundary Refinement	134	82.2%	H	2017	Published	[22]
4	DeepLabv3-JFT	Atrous Convolution in Cascades	Augment ASPP with Image-Level Features	~43	82.7%	M	2017	Preprint	[23]
5	SpineNet-S143+	Scale Permutation and Cross-Scale Connection	NAS Network Discovered from DeepLabv3	162	85.56%	M	2021	Preprint	[24]
6	Eff-B7 NAS-FPN	Simple Copy-Paste Augmentations	Enhanced Dataset Diversity	185	86.6%	L	2020	Published	[25]
7	TADP	Automatically Generated Captions	Model Personalisation	862	87.11%	L	2023	Published	[26]
8	ExFuse	Infusing Semantic Information into Low-Level Features	Introducing High-Resolution Details into High-Level Features	65	87.90%	H	2018	Published	[27]
9	DeepLabv3+	Applying Depthwise Separable Convolution to ASPP and Decoder Module	Introducing a Simple, Effective Decoder Module in DeepLabv3	43	89.0%	M	2018	Published	[28]
10	EfficientNet-L2+NAS-FPN	Self-Training Method	Impact of Data Augmentation and Labeled Data	485	90.5%	L	2020	Published	[29]

Comp. = Comparability level. H (High): evaluated under standard PASCAL VOC 2012 benchmark protocols with no substantial external supervision reported. M (Moderate): includes large-scale pretraining, neural architecture search, or specialised training strategies that may affect direct comparison. L (Limited): relies on self-training, extensive data augmentation, generated captions, diffusion-based representations, large-scale pretraining, or additional external resources that reduce direct comparability with conventional semantic segmentation models. Reported mIoU values are reproduced from the original studies and may not be strictly comparable across different evaluation protocols, training schedules, pretraining datasets, data augmentation strategies, image resolutions, or evaluation settings.

Table 4: Model strategies, parameter size, performance, and comparability level on the ADE20K dataset

SL	Model name	Characteristics 1	Characteristics 2	Parameters (M)	mIoU	Comp.	Year	Source type	Ref
1	DINOv2	Automatic Data Curation Pipeline	Model Distillation	1100	60.2%	L	2021	Preprint	[30]
2	Mask DINO-L	Integration of Mask Prediction Branch	Shared Architecture and Training Process	223	60.8%	M	2022	Published	[31]
3	RevCo-H	Multi-Level Reversible Connection	Versatility and Scalability	2158	61.0%	M	2023	Preprint	[32]
4	FD-SwinV2-G	Feature Distillation	Attention and Optimisation Diagnostic Tools	3000	61.4%	L	2022	Preprint	[33]
5	ViT-Adapter-L	Pretraining-Free Adapter	Leveraging Large-Scale Multi-Modal Data	348	61.5%	L	2023	Preprint	[34]
6	EVA	Pretraining with Masked Image-Text Aligned Vision Features	Vision with Multi-Modal Models	1000	62.3%	L	2022	Published	[35]
7	BEiT-3	Multitask Transformers	Unified Masked "Language" Modeling	1900	62.8%	L	2022	Published	[36]
8	M3F Pre-training	Unified Multi-modal Mutual Information Formula	Single-state Pre-training Pipeline	1000	62.9%	L	2022	Published	[37]
9	InternImage-H	Deformable Convolutions as Core Operator	Reduced Inductive Bias	1800	62.9%	M	2023	Published	[38]
10	ONE-PEACE	Cross-modal Aligning Contrast	Intra-modal Denoising Contrast	1520	63.0%	L	2023	Published	[39]

Comp. = Comparability level. H (High): evaluated under standard ADE20K benchmark protocols with no substantial external supervision reported. M (Moderate): includes specialised architectural designs, feature distillation, or large-scale pretraining that may affect direct comparison. L (Limited): relies on foundation models, multimodal supervision, large-scale image-text pretraining, extensive external data, model distillation, or other external resources that substantially reduce direct comparability with conventional semantic segmentation models. Reported mIoU values are reproduced from the original studies and may not be strictly comparable across different evaluation protocols, training schedules, pretraining datasets, image resolutions, data augmentation strategies, model scales, or evaluation settings.

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